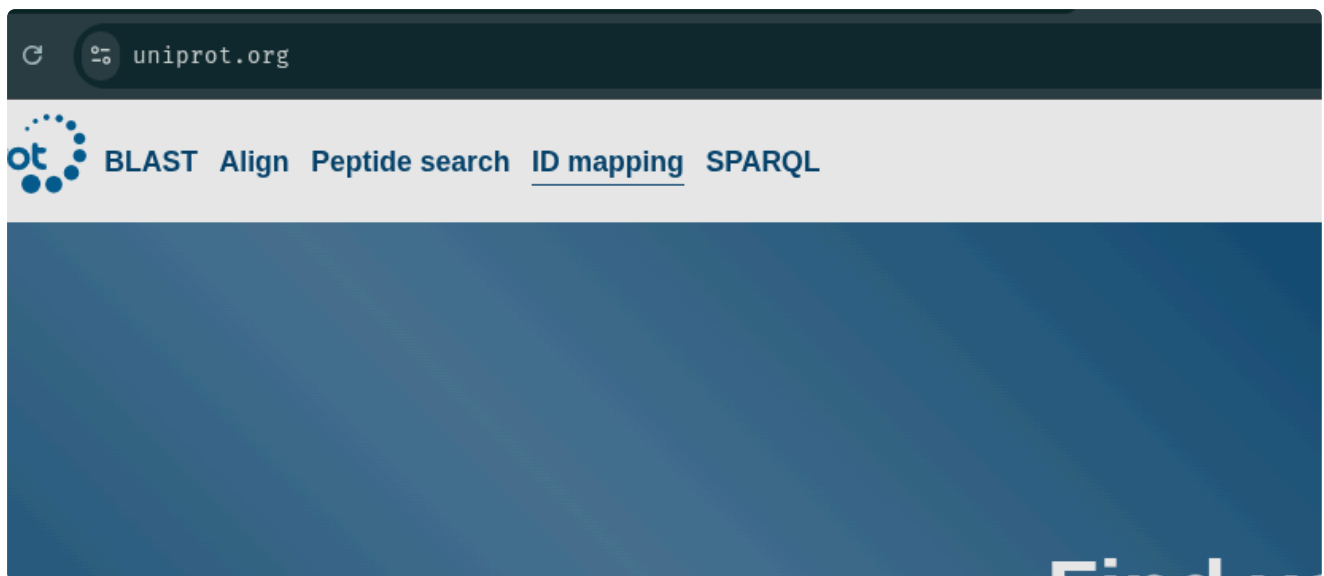


Solution 6

BS24B006

Apoorv Shah

q1



id mapped the set

Retrieve/ID mapping¹

Enter one or more IDs (100,000 max). You may also [load from a text file](#). Separate IDs by whitespace (space, tab, newline) or commas. This service can also be used [programmatically](#). Alternatively, the [underlying data](#) can be downloaded.

P69905, P01946, P01942, P01966, P01958, P01959, P01965, P06635, P60529, P80043, P01980

📘 Your input contains 11 unique IDs

From database: UniProtKB AC/ID ▼ To database: UniProtKB ▼

Name your ID Mapping job

P69905 +10 UniProtKB_AC-ID → UniProtKB


Tools ▾
UniProtKB ▾
Advanced | List
Search




Help

Tool results

Your tool analysis results from the last 7 days are listed below. If you have tools jobs running, you can navigate away to other pages and you will be notified once the job is completed.

ID MAPPING	TPIS_HUMAN +8 UniProtKB_AC-ID → UniProtKB Queued  We will notify you when your results are ready 7w9I8540zr Source database: UniProtKB AC/ID Target database: UniProtKB	2026-09-15 17:28
ID MAPPING	TPIS_HUMAN +8 UniProtKB_AC-ID → UniProtKB Completed 7w9I8540zr Source database: UniProtKB AC/ID Target database: UniProtKB	2026-09-15 17:27
ID MAPPING	P69905 +10 UniProtKB_AC-ID → UniProtKB Completed 233xR1ma3f Source database: UniProtKB AC/ID Target database: UniProtKB	2026-09-15 17:25

Help

mapped the id given in tutorial to get the fasta sequence (given in the folder of repo as well)

Tools Download (11) Add View: Cards Table Customize columns Resubmit Share 11 rows selected

11 IDs mapped to 11 results

✓ From	Entry	Entry Name	Protein Names	Gene Names	Organism	Length
✓ P69905	P69905	HBA_HUMAN	Hemoglobin subunit alpha[...]	HBA1, HBA2	Homo sapiens (Human)	142 AA
✓ P01946	P01946	HBA_RAT	Hemoglobin subunit alpha-1/2 [...]	Hba1, Hba-a1	Rattus norvegicus (Rat)	142 AA
✓ P01942	P01942	HBA_MOUSE	Hemoglobin subunit alpha[...]	Hba, Hba-a1	Mus musculus (Mouse)	142 AA
✓ P01966	P01966	HBA_BOVIN	Hemoglobin subunit alpha[...]	HBA	Bos taurus (Bovine)	142 AA
✓ P01958	P01958	HBA_HORSE	Hemoglobin subunit alpha[...]	HBA	Equus caballus (Horse)	142 AA
✓ P01959	P01959	HBA_EQUAS	Hemoglobin subunit alpha[...]	HBA1, HBA2	Equus asinus (Donkey) (Equus africanus asinus)	142 AA
✓ P01965	P01965	HBA_PIG	Hemoglobin subunit alpha[...]	HBA	Sus scrofa (Pig)	141 AA
✓ P06635	P06635	HBA_PONPY	Hemoglobin subunit alpha[...]	HBA1, HBA2	Pongo pygmaeus (Bornean orangutan)	142 AA
✓ P60529	P60529	HBA_CANLF	Hemoglobin subunit alpha[...]	HBA	Canis lupus familiaris (Dog) (Canis familiaris)	141 AA
✓ P80043	P80043	HBA_TREBE	Hemoglobin subunit alpha[...]	hba	Trematomus bernacchii (Emerald rockcod) (Pseudotrematomus bernacchii)	142 AA
✓ P01980	P01980	HBA_APTFO	Hemoglobin subunit alpha[...]	HBA	Aptenodytes forsteri (Emperor penguin)	142 AA

UniProt Tools Tool results Advanced | List Search Help

Status
Reviewed (Swiss-Prot) (9)

Popular organisms
C. elegans (1)
Fruit fly (1)
Human (1)
Mouse (1)
S. cerevisiae (1)

Taxonomy
Filter by taxonomy

Proteins with
3D structure (6)
Active site (9)
Alternative products (isoforms) (4)
Alternative splicing (4)
Beta strand (6)
More items

Protein existence
Protein level (8)

ID mapping 9 results found for UniProtKB_AC-ID → U

Overview Input Parameters API Request

Tools Download (9) Add View: Cards Table Customize columns Resubmit

9 IDs mapped to 9 results

✓ From	Entry	Entry Name	Protein Names	Gene Names	Organism
✓ TPIS_HUMAN	P60174	TPIS_HUMAN	Triosephosphate isomerase[...]	TPI1, TPI	Homo sapiens (Human)
✓ TPIS_YEAST	P00942	TPIS_YEAST	Triosephosphate isomerase[...]	TPI1, YDR050C, YD9609.05C	Saccharomyces cerevisiae
✓ TPIS_GRAGA	P48492	TPIS_GRAGA	Triosephosphate isomerase, cytosolic [...]	TPI1	Gracilaria (Red algae)
✓ TPIS_TRYCR	P52270	TPIS_TRYCR	Triosephosphate isomerase, glycosomal[...]		Trypanosoma
✓ TPIS_MAIZE	P12863	TPIS_MAIZE	Triosephosphate isomerase, cytosolic [...]		Zea mays

downloaded in fasta

running clustal omega alignment for both the set

GN=hba PE=1 SV=1
SLSDKDKAAVRALWSKIGKSADAIGNDALSRMIVVYPOTKTYFSHWPDVTPGSPHIKAHG
KKVMGGIALAVSKIDDLKTGLMELSEQHAYKL RVD PANFKILNH CILVVISTMFPKEFTP
EAHVSLDKFLSGVALALAERYR

Choose file No file chosen

Close

YOUR JOB IS RUNNING

Please note that results can only be retrieved for jobs submitted within the last seven days.

Job ID: clustalo-I20260915-130143-0172-98497456-p1m

RUNNING

Recent Jobs

Close

sequence More example inputs

Parameters

Submit

The EMBL-EBI Job
tion about this

support staff. If you have
EBI Support. Read our

Privacy Notice if you are concerned with your privacy and how we handle personal information.

EMBL-EBI is the home for big data in biology.

than the specified value. Otherwise, conservation indices will be set to M-S, where M is the mean conservation value and S is the standard deviation.
Default = 0.5

- p Input pdb file [File in] Optional
The sequence in the pdb file should match exactly the first sequence of the alignment.
- d Output pdb file [File Out] Optional
The B-factors are replaced by the conservation indices.
Default = STDOUT

```
> ./al2co -i set1.aln -f 0 -c 0 -n F -o set1_method1_entropy_unweighted.txt
No such file: "set1.aln"
Error with reading alignment: 1
> ./al2co -i setA.aln -f 0 -c 0 -n F -o set1_method1_entropy_unweighted.txt
No such file: "setA.aln"
Error with reading alignment: 1
> ls
al2co  al2co.c  GNUmakefile  README
> cd ..
> ls
al2co  BLOSUM62  SetA.aln  SetA.fasta  SetB.aln  SetB.fasta
> ./al2co -i setA.aln -f 0 -c 0 -n F -o set1_method1_entropy_unweighted.txt
zsh: permission denied: ./al2co
> sudo ./al2co -i setA.aln -f 0 -c 0 -n F -o set1_method1_entropy_unweighted.txt
[sudo] password for specpoorv:
sudo: ./al2co: command not found
> sudo al2co/al2co -i setA.aln -f 0 -c 0 -n F -o set1_method1_entropy_unweighted.txt
No such file: "setA.aln"
Error with reading alignment: 1
> sudo al2co/al2co -i SetA.aln -f 0 -c 0 -n F -o set1_method1_entropy_unweighted.txt
> ls
al2co  BLOSUM62  set1_method1_entropy_unweighted.txt  SetA.aln  SetA.fasta  SetB.aln  SetB.fasta
> al2co/al2co -i SetA.aln -f 0 -c 1 -n F -o set1_method2_variance_unweighted.txt
> al2co/al2co -i SetA.aln -f 0 -c 2 -s BLOSUM62 -n F -o set1_method3_sum_of_pairs.txt
> al2co/al2co -i SetA.aln -f 1 -c 1 -n F -o set1_method4_variance_weighted.txt
> al2co/al2co -i SetA.aln -f 0 -c 0 -n T -o set1_method5_entropy_normalized.txt
> al2co/al2co -i SetB.aln -f 0 -c 0 -n F -o set2_method1_entropy_unweighted.txt
Names do not match, was: 3sp|P52270|TPIS_TRYCR, now: sp|P52270|TPIS_TRYCR
Error with reading alignment: 5
> al2co/al2co -i SetB.aln -f 0 -c 0 -n F -o set2_method1_entropy_unweighted.txt
> al2co/al2co -i SetB.aln -f 0 -c 1 -n F -o set2_method2_variance_unweighted.txt
> al2co/al2co -i SetB.aln -f 0 -c 2 -s BLOSUM62 -n F -o set2_method3_sum_of_pairs.txt
> al2co/al2co -i SetB.aln -f 1 -c 1 -n F -o set2_method4_variance_weighted.txt
> al2co/al2co -i SetB.aln -f 0 -c 0 -n T -o set2_method5_entropy_normalized.txt
~/bioinfo/assignment_6 main !1 ?1 >
```

06:11:14 PM

(for clarification, i have cloned the repo and built it locally)

Q.2

we have to do the same thing except on the web so i will use the same .aln file

Unweighted frequency and entropy-based measure

DATA INPUT

Enter protein sequence alignment in [CLUSTAL](#) format:

Clear sequences

Or upload a file SetA.aln

DATA SUBMIT

Enter **email** to receive the result (optional):

Enter a **job name** (optional):

PARAMETERS

- [sequence weighting scheme](#): ☐ henikoff-henikoff ☐ independent count ☒ unweighted
- [conservation calculation method](#): ☒ entropy ☐ variance ☐ sum-of-pairs
- For sum-of-pairs method only:
[scoring matrix](#):
☒ BLOSUM62 matrix ☐ identity matrix
- [scoring matrix transformation](#):
☒ no transformation ☐ normalization ☐ adjustment
- [normalize conservation values](#): ☐ True ☒ False
- [window size used for averaging conservation \(for smoothing purpose\)](#):
- [gap fraction above which conservation calculation is not performed](#):
- [exclude the first sequence from calculation](#): ☐ True ☒ False
- [output alignment block size](#):
- [pdb file for which b-factor field is replaced with conservation \(optional\)](#):
 No file chosen

[AL2CO documentation](#)

AL2CO alignment conservation server

Your job is number 1 in queue. Please wait ...
This page will be updated automatically every 30 seconds until your job is done.
You can also refresh this page at any time, or bookmark this page and check it later.

RESULTS:

- The list of positional conservation values is [here](#).
- The alignment with integer conservation indices is [here](#).

INPUTS:

- Input alignment is [here](#).
- Input pdb file: none.

http://prodata.swmed.edu/al2co/results/QUERY_3PSvl2.csv.txt

```

1      :      0.000
2      S      -0.305
3      L      0.000
4      S      0.000
5      D      -1.121
6      K      -1.160
7      D      0.000
8      K      0.000
9      A      -1.169
10     A      -0.305
11     V      -0.586
12     R      -0.305
13     A      -1.034
14     L      -1.160
15     W      -0.305
16     S      -0.916
17     K      0.000
18     I      -0.689
19     G      -0.305

```

same as in the file we did through CLI of al2co (all the files present in github)

Unweighted frequency and variance-based measure

Or upload a file
SetA.aln

DATA SUBMIT

Enter **email** to receive the result (optional):
Enter a **job name** (optional):

PARAMETERS

- [sequence weighting scheme](#): ☐ henikoff-henikoff ☐ independent count ☒ unweighted
- [conservation calculation method](#): ☐ entropy ☒ variance ☐ sum-of-pairs
- For sum-of-pairs method only:
[scoring matrix](#):
☒ BLOSUM62 matrix ☐ identity matrix
[scoring matrix transformation](#):
☒ no transformation ☐ normalization ☐ adjustment
- [normalize conservation values](#): ☐ True ☒ False
- [window size used for averaging conservation \(for smoothing purpose\)](#):
- [gap fraction above which conservation calculation is not performed](#):
- [exclude the first sequence from calculation](#): ☐ True ☒ False
- [output alignment block size](#):
- [pdb file for which b-factor field is replaced with conservation \(optional\)](#):
 No file chosen

Unweighted frequency and sum of pairs measure
(HINT: Download and use BLOSUM62 scoring)

matrix)

Or upload a file

Choose file

SetA.aln

DATA SUBMIT

Enter **email** to receive the result (optional):

Enter a **job name** (optional):

Submit

Reset

PARAMETERS

- [sequence weighting scheme](#): ☐ henikoff-henikoff ☐ independent count ☒ unweighted
- [conservation calculation method](#): ☐ entropy ☐ variance ☒ sum-of-pairs
- For sum-of-pairs method only:
[scoring matrix](#):
☒ BLOSUM62 matrix ☐ identity matrix
- [scoring matrix transformation](#):
☒ no transformation ☐ normalization ☐ adjustment
- [normalize conservation values](#): ☐ True ☒ False
- [window size used for averaging conservation \(for smoothing purpose\)](#):
- [gap fraction above which conservation calculation is not performed](#):
- [exclude the first sequence from calculation](#): ☐ True ☒ False
- [output alignment block size](#):
- [pdb file for which b-factor field is replaced with conservation \(optional\)](#):

Choose file

 No file chosen

[AL2CO documentation](#)

Weighted frequency and variance-based measure

Or upload a file SetA.aln

DATA SUBMIT

Enter **email** to receive the result (optional):

Enter a **job name** (optional):

PARAMETERS

- [sequence weighting scheme](#): ☒ henikoff-henikoff ☐ independent count ☐ unweighted
- [conservation calculation method](#): ☐ entropy ☒ variance ☐ sum-of-pairs
- For sum-of-pairs method only:
[scoring matrix](#):
☒ BLOSUM62 matrix ☐ identity matrix
- [scoring matrix transformation](#):
☒ no transformation ☐ normalization ☐ adjustment
- [normalize conservation values](#): ☐ True ☒ False
- [window size used for averaging conservation \(for smoothing purpose\)](#):
- [gap fraction above which conservation calculation is not performed](#):
- [exclude the first sequence from calculation](#): ☐ True ☒ False
- [output alignment block size](#):
- [pdb file for which b-factor field is replaced with conservation \(optional\)](#):
 No file chosen

[AL2CO documentation](#)

Normalize the scores obtained with (i)

Or upload a file SetA.aln

DATA SUBMIT

Enter **email** to receive the result (optional):

Enter a **job name** (optional):

PARAMETERS

- [sequence weighting scheme](#): ☐ henikoff-henikoff ☐ independent count ☒ unweighted
- [conservation calculation method](#): ☒ entropy ☐ variance ☐ sum-of-pairs
- For sum-of-pairs method only:
[scoring matrix](#):
☒ BLOSUM62 matrix ☐ identity matrix
- [scoring matrix transformation](#):
☒ no transformation ☐ normalization ☐ adjustment
- [normalize conservation values](#): ☒ True ☐ False
- [window size used for averaging conservation \(for smoothing purpose\)](#):
- [gap fraction above which conservation calculation is not performed](#):
- [exclude the first sequence from calculation](#): ☐ True ☒ False
- [output alignment block size](#):
- [pdb file for which b-factor field is replaced with conservation \(optional\)](#):
 No file chosen

```
1      :      0.943
2      S      0.253
3      L      0.943
4      S      0.943
5      D     -1.597
6      K     -1.685
7      D      0.943
8      K      0.943
9      A     -1.705
10     A      0.253
11     V     -0.385
12     R      0.253
13     A     -1.399
14     L     -1.685
15     W      0.253
16     S     -1.134
17     K      0.943
18     I     -0.618
19     G      0.253
```

```
assignment_07 = seq1_method5_entropy_1
1      1      -      0.943
2      2      S      0.253
3      3      L      0.943
4      4      S      0.943
5      5      D     -1.597
6      6      K     -1.685
7      7      D      0.943
8      8      K      0.943
9      9      A     -1.705
10     10     A      0.253
11     11     V     -0.385
12     12     R      0.253
13     13     A     -1.399
```

both files show same value and it looks normalised as well with respect to 1

q.3

```
assignment_6 > get_top_residues.py > ...
5 def parse_al2co_file(filepath):
13     parts = line.split()
14     if len(parts) >= 3 and parts[0].isdigit():
15         pos = int(parts[0])
16         res = parts[1]
17         if "*" in parts or parts[2].endswith("*"):
18             continue
19
20         score_str = parts[2].replace("*", "")
21         try:
22             score = float(score_str)
23             data.append((pos, res, score))
24         except ValueError:
25             continue
26     return data
27
28 # i wrote this
29 def print_top_conservation(data, n=10):
30     sorted_highest = sorted(data, key=lambda x: x[2], reverse=True)[:n]
31     sorted_lowest = sorted(data, key=lambda x: x[2])[:n]
32
33     print(f"Top {n} Highest Conservation Scores")
34     print(f"{'Rank':<6}{'Position':<10}{'Residue':<10}{'Score':<10}")
35     for rank, (pos, res, score) in enumerate(sorted_highest, start=1):
36         print(f"{rank:<6}{pos:<10}{res:<10}{score:<10.3f}")
37
38     print(f"Top {n} Lowest Conservation Scores")
39     print(f"{'Rank':<6}{'Position':<10}{'Residue':<10}{'Score':<10}")
40     for rank, (pos, res, score) in enumerate(sorted_lowest, start=1):
41         print(f"{rank:<6}{pos:<10}{res:<10}{score:<10.3f}")
42
43
44 if __name__ == "__main__":
45     filename = (
46         sys.argv[1]
47         if len(sys.argv) > 1
48         else "/home/specapoorv/bioinfo/assignment_6/set1_method1_entropy_unweighted.txt"
49     )
50     records = parse_al2co_file(filename)
51     print_top_conservation(records, n=10)
```

wrote this thingy (file in github again)

```
FileNotFoundError: [Errno 2] No such file or directory: 'set1_method1_entropy_unweighted
>
> /usr/bin/python3 /home/specapoorv/bioinfo/assignment_6/get_top_residues.py
Top 10 Highest Conservation Scores
Rank  Position  Residue  Score
1      1         -      0.000
2      3         L      0.000
3      4         S      0.000
4      7         D      0.000
5      8         K      0.000
6     17         K      0.000
7     26         G      0.000
8     30         L      0.000
9     32         R      0.000
10    38         P      0.000
Top 10 Lowest Conservation Scores
Rank  Position  Residue  Score
1    117         K     -1.846
2    113         T     -1.720
3     70         L     -1.673
4     73         S     -1.594
5     84         E     -1.414
6    133         G     -1.414
7     36         V     -1.367
8     35         V     -1.295
9      9         A     -1.169
10     6         K     -1.160
~/bioinfo main !1 ?1 > |
```

Set A

Rank	Alignment Position	Residue	Conservation Score (Entropy)
1	100	F	0.000
2	101	K	0.000
3	103	L	0.000
4	116	P	0.000
5	121	P	0.000
6	124	H	0.000
7	126	S	0.000
8	127	L	0.000
9	128	D	0.000
10	129	K	0.000

Rank	Alignment Position	Residue	Conservation Score (Entropy)
1	117	K	-1.846
2	113	T	-1.720
3	70	L	-1.673
4	73	S	-1.594
5	133	G	-1.414
6	84	E	-1.414
7	36	V	-1.367
8	35	V	-1.295
9	9	A	-1.169
10	14	L	-1.160

SetB

Rank	Alignment Position	Residue	Conservation Score (Entropy)
1	117	K	-1.846
2	113	T	-1.720
3	70	L	-1.673
4	73	S	-1.594
5	133	G	-1.414
6	84	E	-1.414
7	36	V	-1.367
8	35	V	-1.295
9	9	A	-1.169
10	14	L	-1.160

Rank	Alignment Position	Residue	Conservation Score (Entropy)
1	33	T	-1.889
2	64	Q	-1.831
3	147	A	-1.831
4	27	E	-1.735
5	23	V	-1.677
6	36	H	-1.677
7	145	R	-1.677
8	196	L	-1.677
9	199	R	-1.677
10	208	D	-1.677

q.4

core logic is this

```
print(f"{'Pos':<6}{'Entropy':<12}{'Variance':<12}{'SumOfPairs':<12}")

for i in range(aln_len):
    col = [s[i].upper() for s in seqs]

    #ignore gaps to count frequencies
    valid_aas = [aa for aa in col if aa in AMINO_ACIDS]

    if not valid_aas:
        continue

    #unweighted frequencies: count / total_valid
    freqs = {aa: valid_aas.count(aa) / len(valid_aas) for aa in AMINO_ACIDS}

    entropy_score = calc_entropy(freqs)
    variance_score = calc_variance(freqs)
    sop_score = calc_sum_of_pairs(freqs, matrix)

    print(
        f"{i+1:<6}{entropy_score:<12.3}{variance_score:<12.3}{sop_score:<12.3}"
    )
```

we first calculate the unweighted frequencies and pass them into this function

```

def calc_entropy(freqs):
    #C(i) = sum( f_a * ln(f_a) )
    score = 0.0
    for aa in AMINO_ACIDS:
        f = freqs[aa]
        if f > 0:
            score += f * math.log(f)
    return score

def calc_variance(freqs, bg_freq=1.0 / 20.0):
    # C(i) = sqrt( sum( (f_a - f_overall)^2 ) )
    sq_diff_sum = sum((freqs[aa] - bg_freq) ** 2 for aa in AMINO_ACIDS)
    return math.sqrt(sq_diff_sum)

def calc_sum_of_pairs(freqs, matrix):
    # C(i) = sum_a sum_b ( f_a * f_b * S_ab )
    score = 0.0
    for a in AMINO_ACIDS:
        for b in AMINO_ACIDS:
            if freqs[a] > 0 and freqs[b] > 0:
                score += freqs[a] * freqs[b] * matrix.get((a, b), 0)
    return score

```

matrix used here is blosum62 matrix

```

91
92
93 def main():
94     aln_file = "SetA.aln"
95     blosum_file = "BLOSUM62"
96

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

•> cd assignment_6
•>
•> /usr/bin/python3 /home/specapoorv/bioinfo/assignment_6/q4.py

```

Pos	Entropy	Variance	SumOfPairs
1	0.0	0.975	5.0
2	-0.305	0.886	3.01
3	0.0	0.975	4.0
4	0.0	0.975	4.0
5	-1.12	0.582	1.15
6	-1.16	0.623	1.16
7	0.0	0.975	6.0
8	0.0	0.975	5.0
9	-1.17	0.567	1.87
10	-0.305	0.886	4.66
11	-0.586	0.744	3.6
12	-0.305	0.886	4.5
13	-1.03	0.636	1.91
14	-1.16	0.623	1.52
15	-0.305	0.886	9.31
16	-0.916	0.623	2.26
17	0.0	0.975	5.0
18	-0.689	0.674	3.5
19	-0.305	0.886	4.99
20	-1.03	0.636	2.4
21	-0.886	0.71	4.37
22	-0.474	0.808	2.88
23	-1.03	0.636	2.24
24	-0.908	0.649	2.45
25	-0.886	0.71	4.31
26	0.0	0.975	6.0
27	-0.886	0.71	1.83
28	-0.305	0.886	4.51
29	-0.305	0.886	3.35

the answer matches with the files from al2co with some slight difference maybe because i did it for only 2 decimals

q.6

MAFFT < Job Di x +

ebi.ac.uk/jdispatcher/msa/mafft

New Chrome available

Important note: This tool can align up to 500 sequences or a maximum file size of 1 MB.

Input sequence ⓘ

Sequence type

☐ Automatic ☒ Protein ☐ DNA

Parameters

Submit

More example inputs

Close

YOUR JOB IS SUBMITTING

Please note that results can only be retrieved for jobs submitted within the last seven days.

Job ID: mafft-I20260921-165603-0435-94787005-p1m

SUBMITTING

Recent Jobs

Close

MAFFT
Multiple Sequence Alignment (MSA)

Home Help & Privacy Recent Jobs **Input Form** [Feedback](#)

Results for Job ID **mafft-I20260921-165603-0435-94787005-p1m** [Copy](#) [Resubmission](#)

< **Tool Output** Alignments Guide Tree Phylogenetic Tree Results Viewers Result Files >

Tool output [Download](#)

```
>sp|P01942|HBA_MOUSE Hemoglobin subunit alpha OS=Mus musculus OX=10090
GN=Hba PE=1 SV=2
MVLSGEDKSNIAAWGKIGGHGAEYGAEALERMFAFPTTKTYFPHF-DVSHGSAQVKGH
GKKVADALASAAGHLDLPGALSALSDLHAHKLRVDPVNFKLLSHCLLVTLASHHPADFT
PAVHASLDKFLASVSTVLTISKYR
>sp|P01958|HBA_HORSE Hemoglobin subunit alpha OS=Equus caballus OX=9796
GN=HBA PE=1 SV=2
MVLSAADKTNVKAWSKVGHGAGEYGAEALERMFLGFPTTKTYFPHF-DLSHGSAQVKAH
GKKVGDAITLAVGHLDLPGALSNLSDLHAHKLRVDPVNFKLLSHCLLSTLAVHLPNDFT
PAVHASLDKFLSSVSTVLTISKYR
>sp|P01959|HBA_EQUAS Hemoglobin subunit alpha OS=Equus asinus OX=9793
GN=HBA1 PE=1 SV=3
MVLSTADKTNVKAWSKVGHGAGEYGAEALERMFLGFPTTKTYFPHF-DLSHGSAQVKAH
GKKVGDAITLAVGHLDLPGALSNLSDLHAHKLRVDPVNFKLLSHCLLSTLAVHLPNDFT
PAVHASLDKFLSSVSTVLTISKYR
```

got mafft and muscle .aln files for both the sets

since i used al2co and then parsed them to get top residues i ran al2co again for these 6 files with method 1

```

> ./al2co -i setA-mafft.aln -f 0 -c 0 -n F -o set1_mafft_entropy.txt
zsh: permission denied: ./al2co
> al2co/al2co -i setA-mafft.aln -f 0 -c 0 -n F -o set1_mafft_entropy.txt
No such file: "setA-mafft.aln"
Error with reading alignment: 1
> ls
al2co    q3.py    q5.py    SetA.aln  SetA-mafft.aln  SetB.aln  SetB-mafft.aln
BLOSUM62 q4.py    results  SetA.fasta  SetA-Muscle.aln  SetB.fasta  SetB-Muscle.aln
> ls /dev/input/js*
> al2co/al2co -i SetA-mafft.aln -f 0 -c 0 -n F -o set1_mafft_entropy.txt
wrong len for MVLSGEDKSNIAAWGKIGGHGAEYGAEALERMFAFPTTKTYFPHF-DVSHGSAQVKGH, was: 10, now: 0
Error with reading alignment: 6
~/bioinfo/assignment_6 main !12 ?8 >

```

it gave some error, because mafft on default gives in fasta
so i re ran to get it as clustalw format instead

☐ Automatic
 ☒ Protein
 ☐ DNA

Paste your sequence here - or use the example sequence

Choose file **SetA.fasta**

Use the example

Clear sequence

More example inputs

Parameters

OUTPUT FORMAT ⓘ

ClustalW
▼

More options ▼

Submit

Title

MAFFT's job

Submit

runs correctly now

```

ls /dev/input/js*
al2co/al2co -i SetA-mafft.aln -f 0 -c 0 -n F -o set1_mafft_entropy.txt
wrong len for MVLSGEDKSNIAAWGKIGGHGAIEYGAELERMFASPTTKTYFPHF-DVSHGSAQVKGH, was: 10, now: 0
Error with reading alignment: 6
al2co/al2co -i SetA-Muscle.aln -f 0 -c 0 -n F -o set1_muscle_entropy.txt
al2co/al2co -i SetA-mafft.aln -f 0 -c 0 -n F -o set1_mafft_entropy.txt
~/bioinfo/assignment_6 main !12 ?10 >
  
```

09:46:39 PM

```

> /usr/bin/python3 /home/specapoorv/bioinfo/assignment_6/q5.py
=== (i) Top Residues with SIMILAR Conservation Scores (Diff <= 0.1) ===
Pos  Res  Clustal  MAFFT  MUSCLE  Diff
1    -    0.000    0.000    0.000    0.000
2    S    -0.305   -0.305   -0.305    0.000
3    L    0.000    0.000    0.000    0.000
4    S    0.000    0.000    0.000    0.000
5    D    -1.121   -1.121   -1.121    0.000
6    K    -1.160   -1.160   -1.160    0.000
7    D    0.000    0.000    0.000    0.000
8    K    0.000    0.000    0.000    0.000
9    A    -1.169   -1.169   -1.169    0.000
10   A    -0.305   -0.305   -0.305    0.000

=== (ii) Top Residues with DIFFERENT Conservation Scores (Diff > 0.1) ===
Pos  Res  Clustal  MAFFT  MUSCLE  Diff
>
> /usr/bin/python3 /home/specapoorv/bioinfo/assignment_6/q5.py
=== (i) Top Residues with SIMILAR Conservation Scores (Diff <= 0.1) ===
Pos  Res  Clustal  MAFFT  MUSCLE  Diff
3    S    -0.974   -0.974   -0.974    0.000
4    K    -1.494   -1.494   -1.494    0.000
5    P    -0.377   -0.377   -0.377    0.000
6    Q    -0.736   -0.736   -0.736    0.000
7    P    -0.377   -0.377   -0.377    0.000
8    I    -0.736   -0.736   -0.736    0.000
9    A    -0.377   -0.377   -0.377    0.000
10   A    -0.377   -0.377   -0.377    0.000
11   A    -0.377   -0.377   -0.377    0.000
12   N    0.000    0.000    0.000    0.000

=== (ii) Top Residues with DIFFERENT Conservation Scores (Diff > 0.1) ===
Pos  Res  Clustal  MAFFT  MUSCLE  Diff
126  A    -1.061   -1.061   -0.349    0.712
71   I    -0.849   -0.562   -0.562    0.287
~/bioinfo/main 112 213 >

```

top one is of set1 or setA, bottom is of SetB

so basically For Set 1, all three aligners (Clustal Omega, MAFFT, and MUSCLE) produced the exact same alignment with zero score difference across all residues because the globin sequences share very high sequence similarity with no tricky insertions.

For Set 2, most core residues still matched completely (diff = 0.000), but a few positions like 126 (diff = 0.712) and 71 (diff = 0.287) had different conservation scores. This happens because the sequences in Set 2 are from widely different species with variable loop regions, so MUSCLE placed gaps and shifted columns differently compared to Clustal and MAFFT

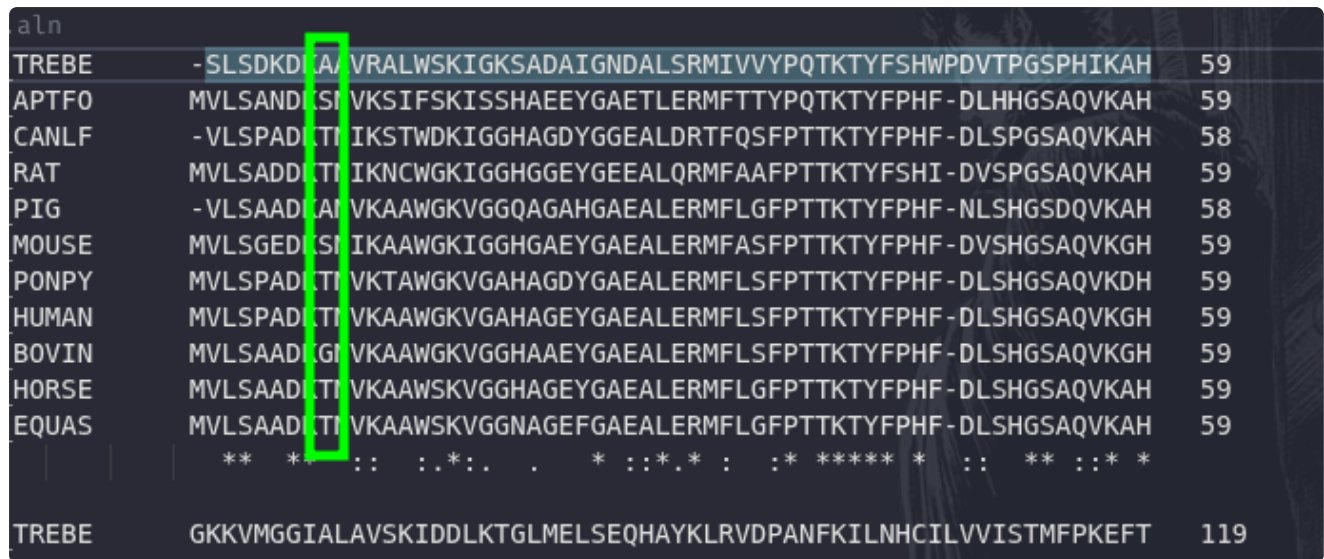
q.6

entropy formula is

$$C(i) = \sum_a f_a \ln(f_a)$$

so we will go at each position count the AAs and calculate frequencies
and for each residue at that position calculate the entropy

position 9



The image shows a sequence alignment of various proteins. A green box highlights the 9th column of the alignment, which contains the residues: A, S, T, T, A, S, T, V, A, I, V, A. Below the alignment, a conservation score is shown for this position: ** * : : . : * : * : * : * : * : * : *

Sequence	Position 9	Length
TREBE	A	59
APTFO	S	59
CANLF	T	58
RAT	T	59
PIG	A	58
MOUSE	S	59
PONPY	T	59
HUMAN	V	59
BOVIN	A	59
HORSE	I	59
EQUAS	V	59

count:

A = 2

S = 2

T = 6

G = 1

Frequencies:

- $f_A = \frac{2}{11} \approx 0.1818$
- $f_S = \frac{2}{11} \approx 0.1818$
- $f_T = \frac{6}{11} \approx 0.5455$
- $f_G = \frac{1}{11} \approx 0.0909$

$$f_A \ln(f_A) = 0.1818 \times \ln(0.1818) = 0.1818 \times (-1.7048) \approx -0.3099$$

$$f_S \ln(f_S) = 0.1818 \times \ln(0.1818) = 0.1818 \times (-1.7048) \approx -0.3099$$

$$f_T \ln(f_T) = 0.5455 \times \ln(0.5455) = 0.5455 \times (-0.6061) \approx -0.3306$$

$$f_G \ln(f_G) = 0.0909 \times \ln(0.0909) = 0.0909 \times (-2.3979) \approx -0.2180$$

$$C(9) = -0.3099 - 0.3099 - 0.3306 - 0.2180 = -1.1684 \approx -1.169$$

pos - 11

sp P80043 HBA_TREBE	-SLSDKDKAAV	ALWSKIGKSADAIGNDALSRMIVVYPQTKTYFSHWPDVTPGSPHIKAH	59
sp P01980 HBA_APTFO	MVLSANDKS	IVLSIFSISKISSHAEYGAETLERMFTTYPQTKTYFPHF-DLHHGSAQVKAH	59
sp P60529 HBA_CANLF	-VLSPADKT	IIISTWDKIGGHAGDYGGGEALDRTFQSFPTTKTYFPHF-DLSPGSAQVKAH	58
sp P01946 HBA_RAT	MVLSADDKT	IIINCWGKIGGHGGEYGEEALQRMFAAFPTTKTYFSHI-DVSPGSAQVKAH	59
sp P01965 HBA_PIG	-VLSAADKA	IVIAAWGKVGGAAGAHGAEALERMFLGFPTTKTYFPHF-NLSHGSDQVKAH	58
sp P01942 HBA_MOUSE	MVLSGEDKS	IIIAAWGKIGGHGAEYGAEALERMFLGFPTTKTYFPHF-DVSHGSAQVKGH	59
sp P06635 HBA_PONPY	MVLSPADKT	IVITAWGKVGAGHAGDYGAEALERMFLSFPTTKTYFPHF-DLSHGSAQVKDH	59
sp P69905 HBA_HUMAN	MVLSPADKT	IVIAAWGKVGAGHAGEYGAEALERMFLSFPTTKTYFPHF-DLSHGSAQVKGH	59
sp P01966 HBA_BOVIN	MVLSAADKG	IVIAAWGKVGGAHAEYGAEALERMFLSFPTTKTYFPHF-DLSHGSAQVKGH	59
sp P01958 HBA_HORSE	MVLSAADKT	IVIAAWSKVGGHAGEYGAEALERMFLGFPTTKTYFPHF-DLSHGSAQVKAH	59
sp P01959 HBA_EQUAS	MVLSAADKT	IVIAAWSKVGGNAGEFGAEALERMFLGFPTTKTYFPHF-DLSHGSAQVKAH	59

count:

$$V = 8$$

$$I = 3$$

- Frequencies:**

- $$f_V = \frac{8}{11} \approx 0.7273$$

- $$f_I = \frac{3}{11} \approx 0.2727$$

- Entropy calculation:**

-

$$f_V \ln(f_V) = 0.7273 \times \ln(0.7273) = 0.7273 \times (-0.3184) \approx -0.2316$$

$$f_I \ln(f_I) = 0.2727 \times \ln(0.2727) = 0.2727 \times (-1.2993) \approx -0.3543$$

$$C(11) = -0.2316 - 0.3543 = -0.5859 \approx -0.586$$

pos -20

sp P80043 HBA_TREBE	-SLSDKDKAAV	RALWSKIGKSADAIGNDALSRMIVVYPQTKTYFSHWPDVTPGSPHIKAH	59
sp P01980 HBA_APTFO	MVLSANDKSNVKS	IFSISKISSHAEYGAETLERMFTTYPQTKTYFPHF-DLHHGSAQVKAH	59
sp P60529 HBA_CANLF	-VLSPADKTNIK	STWDKIGGHAGDYGGGEALDRTFQSFPTTKTYFPHF-DLSPGSAQVKAH	58
sp P01946 HBA_RAT	MVLSADDKTNIK	NCWGKIGGHGGEYGEEALQRMFAAFPTTKTYFSHI-DVSPGSAQVKAH	59
sp P01965 HBA_PIG	-VLSAADKANV	KAAWGKVGGAAGAHGAEALERMFLGFPTTKTYFPHF-NLSHGSDQVKAH	58
sp P01942 HBA_MOUSE	MVLSGEDKSN	IKAAWGKIGGHGAEYGAEALERMFLGFPTTKTYFPHF-DVSHGSAQVKGH	59
sp P06635 HBA_PONPY	MVLSPADKTNV	KTAWGKVGAGHAGDYGAEALERMFLSFPTTKTYFPHF-DLSHGSAQVKDH	59
sp P69905 HBA_HUMAN	MVLSPADKTNV	KAAWGKVGAGHAGEYGAEALERMFLSFPTTKTYFPHF-DLSHGSAQVKGH	59
sp P01966 HBA_BOVIN	MVLSAADKGNV	KAAWGKVGGAHAEYGAEALERMFLSFPTTKTYFPHF-DLSHGSAQVKGH	59
sp P01958 HBA_HORSE	MVLSAADKTNV	KAAWSKVGGHAGEYGAEALERMFLGFPTTKTYFPHF-DLSHGSAQVKAH	59
sp P01959 HBA_EQUAS	MVLSAADKTNV	KAAWSKVGGNAGEFGAEALERMFLGFPTTKTYFPHF-DLSHGSAQVKAH	59

Count:

$$K = 1$$

S = 1

G = 7

A = 2

- **Frequencies:**

- $f_K = \frac{1}{11} \approx 0.0909$

- $f_S = \frac{1}{11} \approx 0.0909$

- $f_G = \frac{7}{11} \approx 0.6364$

- $f_A = \frac{2}{11} \approx 0.1818$

- ****Entropy calculation:**

$$f_K \ln(f_K) = 0.0909 \times \ln(0.0909) \approx -0.2180$$

$$f_S \ln(f_S) = 0.0909 \times \ln(0.0909) \approx -0.2180$$

$$f_G \ln(f_G) = 0.6364 \times \ln(0.6364) = 0.6364 \times (-0.45198) \approx -0.2876$$

$$f_A \ln(f_A) = 0.1818 \times \ln(0.1818) \approx -0.3099$$

$$C(20) = -0.2180 - 0.2180 - 0.2876 - 0.3099 = -1.0335 \approx -1.034$$

pos-22

sp P80043 HBA_TREBE	-SLSDKDKAAVRALWSKIGKSAIAIGNDALSRMIVVYPQTKTYFSHWPDVTPGSPHIKAH	59
sp P01980 HBA_APTFO	MVLSANDKSNVKSIFSKEISSIAIEYGAETLERMFTTYPQTKTYFPHF-DLHHGSAQVKAH	59
sp P60529 HBA_CANLF	-VLSPADKTNIKSTWDKIGGIAQDYGGAEALDRTFQSFPPTTKTYFPHF-DLSPGSAQVKAH	58
sp P01946 HBA_RAT	MVLSADDKTNKNCWGKIGGIAQDYGGAEALQRMFAAFPTTKTYFSHI-DVSPGSAQVKAH	59
sp P01965 HBA_PIG	-VLSAADKANVKAAGKVGGAQAHGAEALERMFSGFPPTTKTYFPHF-NLSHGSDQVKAH	58
sp P01942 HBA_MOUSE	MVLSGEDKSNIAAAGKIGGIAQDYGGAEALERMFASFPPTTKTYFPHF-DVSHGSAQVKGH	59
sp P06635 HBA_PONPY	MVLSPADKTNVKTAWGKVGAIAQDYGAEALERMFSLFPPTTKTYFPHF-DLSHGSAQVKDH	59
sp P69905 HBA_HUMAN	MVLSPADKTNVKAAGKVGGAIAQDYGAEALERMFSLFPPTTKTYFPHF-DLSHGSAQVKGH	59
sp P01966 HBA_BOVIN	MVLSAADKGNVKAAGKVGGAIAQDYGAEALERMFSLFPPTTKTYFPHF-DLSHGSAQVKGH	59
sp P01958 HBA_HORSE	MVLSAADKTNVKAWSKVGGIAQDYGAEALERMFSGFPPTTKTYFPHF-DLSHGSAQVKAH	59
sp P01959 HBA_EQUAS	MVLSAADKTNVKAWSKVGGIAQDYGAEALERMFSGFPPTTKTYFPHF-DLSHGSAQVKAH	59
	** ** :: :*. * * :*. * : * ***** * :: ** :* *	

count:

A = 9

G = 2

- **Frequencies:**

- $f_A = \frac{9}{11} \approx 0.8182$

- $f_G = \frac{2}{11} \approx 0.1818$

- **Entropy calculation:**

$$f_A \ln(f_A) = 0.8182 \times \ln(0.8182) = 0.8182 \times (-0.2007) \approx -0.1642$$

$$f_G \ln(f_G) = 0.1818 \times \ln(0.1818) = 0.1818 \times (-1.7048) \approx -0.3099$$

$$C(22) = -0.1642 - 0.3099 = -\mathbf{0.4741} \approx -\mathbf{0.474}$$

pos-30

sp P80043 HBA_TREBE	-SLSDKDKAAVRALWSKIGKSADAIGNDALE	RMIVVYPQTKTYFSHPDVTGSPHIKAH	59
sp P01980 HBA_APTFO	MVLSANDKSNVKSIFSISKISSHAEYGAETLE	RMFTTYPQTKTYFPHF-DLHHGSAQVKAH	59
sp P60529 HBA_CANLF	-VLSPADKTNIKSTWDKIGGHAGDYGGEALE	RTFQSFPPTTKTYFPHF-DLSPGSAQVKAH	58
sp P01946 HBA_RAT	MVLSADDKTNIKNCWKGKIGGHGGEYGEELC	RMFAAFPTTKTYFSHI-DVSPGSAQVKAH	59
sp P01965 HBA_PIG	-VLSAADKANVKAAGKVGQAGAHGAELER	MFSGFPTTKTYFPHF-NLSHGSDQVKAH	58
sp P01942 HBA_MOUSE	MVLSGEDKSNIKAAWGKIGGHGAEYGAELER	MFASFPTTKTYFPHF-DVSHGSAQVKGH	59
sp P06635 HBA_PONPY	MVLSPADKTNVKTAWGKVGGAHAGDYGAELER	MFSLFPTTKTYFPHF-DLSHGSAQVKDH	59
sp P69905 HBA_HUMAN	MVLSPADKTNVKAAGKVGGAHAGEYGAELER	MFSLFPTTKTYFPHF-DLSHGSAQVKGH	59
sp P01966 HBA_BOVIN	MVLSAADKGNVKAAGKVGGAHAEYGAELER	MFSLFPTTKTYFPHF-DLSHGSAQVKGH	59
sp P01958 HBA_HORSE	MVLSAADKTNVKAWSKVGGAHAGEYGAELER	MFSGFPTTKTYFPHF-DLSHGSAQVKAH	59
sp P01959 HBA_EQUAS	MVLSAADKTNVKAWSKVGGNAGEFGAELER	MFSGFPTTKTYFPHF-DLSHGSAQVKAH	59
	** ** :: :.*. . * : ** : : * ***** * :: ** :.* *		

count :

L = 11

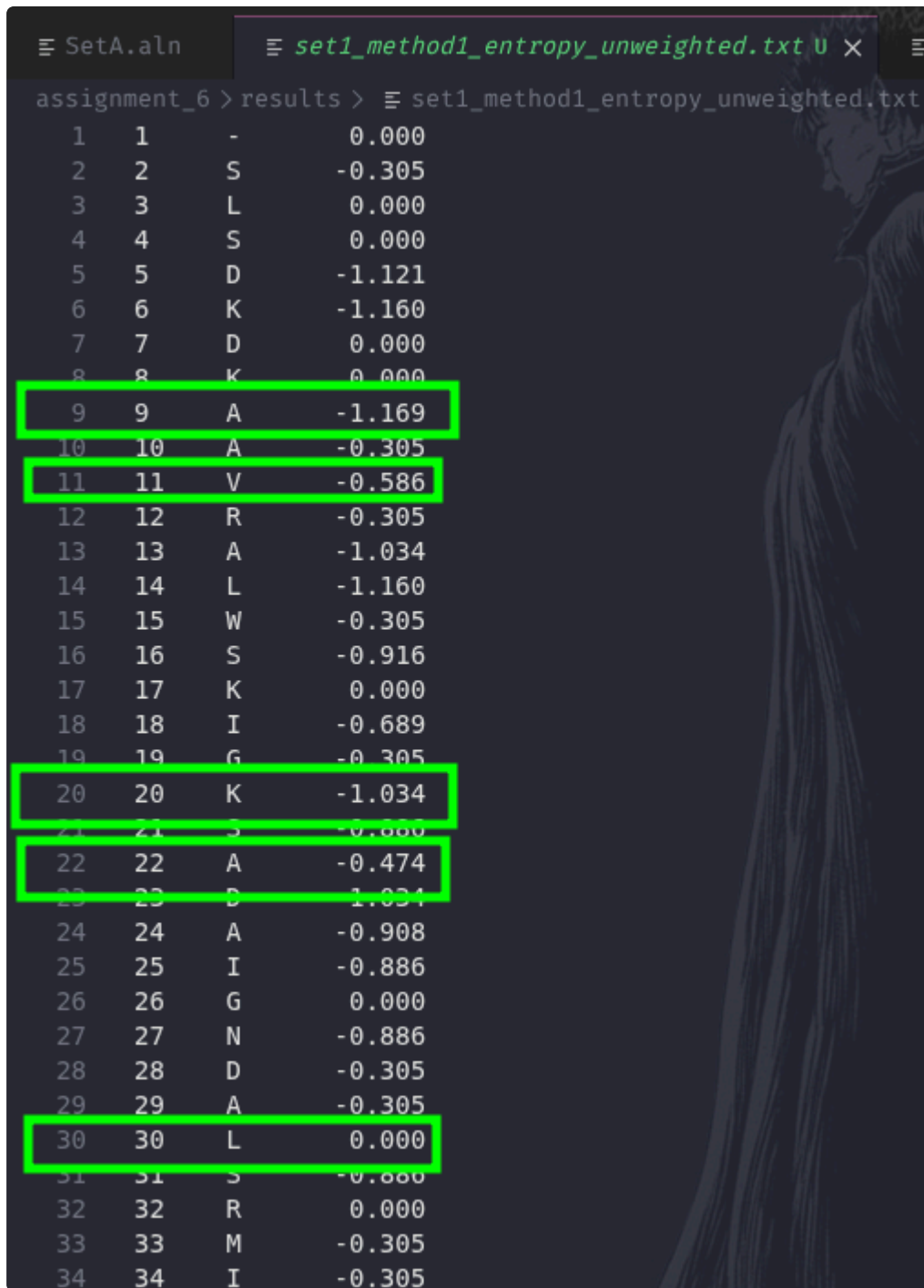
- **Frequencies:**

- $f_L = \frac{11}{11} = 1.0$

- **Entropy calculation:**

$$C(30) = 1.0 \times \ln(1.0) = 1.0 \times 0 = \mathbf{0.000}$$

Position	Manual Calculated Score	AL2CO Output Score
9	-1.169	-1.169
11	-0.586	-0.586
20	-1.034	-1.034
22	-0.474	-0.474
30	0.000	0.000



assignment_6 > results > set1_method1_entropy_unweighted.txt

1	1	-	0.000
2	2	S	-0.305
3	3	L	0.000
4	4	S	0.000
5	5	D	-1.121
6	6	K	-1.160
7	7	D	0.000
8	8	K	0.000
9	9	A	-1.169
10	10	A	-0.305
11	11	V	-0.586
12	12	R	-0.305
13	13	A	-1.034
14	14	L	-1.160
15	15	W	-0.305
16	16	S	-0.916
17	17	K	0.000
18	18	I	-0.689
19	19	G	-0.305
20	20	K	-1.034
21	21	S	-0.886
22	22	A	-0.474
23	23	D	-1.034
24	24	A	-0.908
25	25	I	-0.886
26	26	G	0.000
27	27	N	-0.886
28	28	D	-0.305
29	29	A	-0.305
30	30	L	0.000
31	31	S	-0.886
32	32	R	0.000
33	33	M	-0.305
34	34	I	-0.305

q.7

☒ Amino Acids ☐ Nucleotides

Is there a known structure? 

Insert PDB/UniProt ID

1BTM

or upload a structure 

Select chain identifier

Chain A

☐

No, there is no known structure

Insert a sequence

Paste your sequence here fasta format only

Or, upload sequence File 

If you want to predict the 3D structure using MODELLER, insert MODELLER key 

Job name **required*

1BTM_A

Your email address **required*

apoorvshah024@gmail.com

Submit With Default Parameters

Clear

Select Run Parameters Manually

submitted with default parameters, i dont understand most of the parameters listed there



ConSurf Job Status Page - **FINISHED**

[Go To The Results](#)

Job title

1BTM_A

Date

21/09/2026

Running Parameters

Structure

PDB ID: 1BTM

Chain identifier: A

Phylogenetic Progress

Neighbor Joining with ML
distance

Alignment

Multiple Sequence Alignment is
built using MAFFT

The Homologues are collected
from UNIREF90

Homolog search algorithm:
HMMER

HMMER E-value: 0.0001

No. of HMMER Iterations: 1

Maximal %ID Between
Sequences : 95

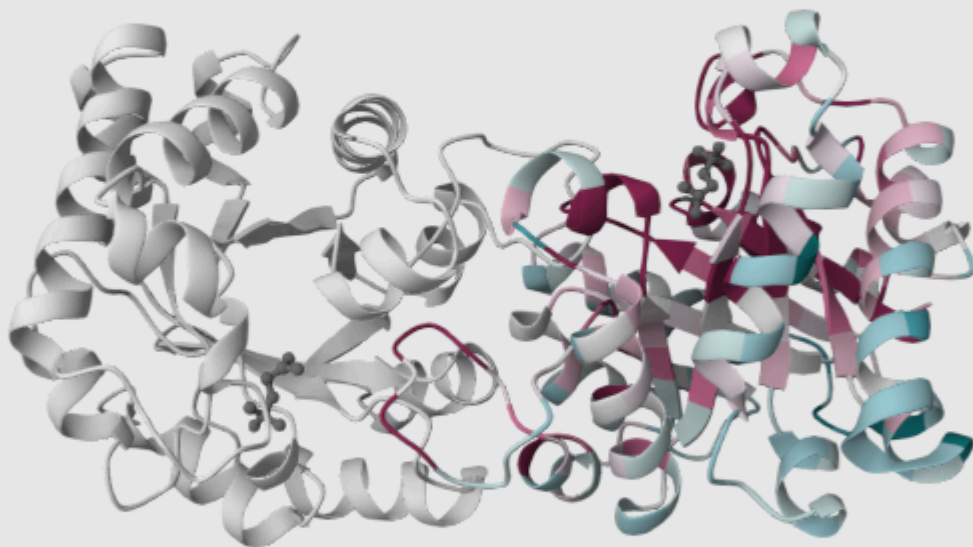
Minimal %ID For Homologs : 35

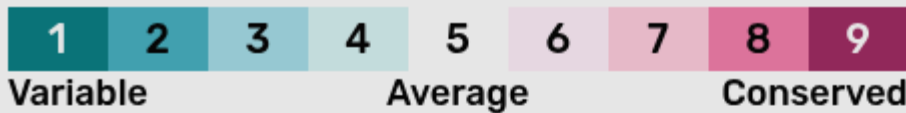
Sequence of 1BTM Chain 1: TRIOSEPH... A

1 RKPIIAGNWKMHKTLAEAVQFVEDVKGHVPPADEVISVVCAPFLFLDRLVQAADGTDLKIGAQTMHFADQGAYTGEVSP
81 11 21 31 41 51 61 71
VMLKDLGVTYVILGHSERRQMFAETDETVNKKVLAATRGLIPIICCGESLEEREAGQTNAVVASQVEKALAGLTPEQV
161 91 101 111 121 131 141 151
KQAVIAYEPIWAIGTGSSTPEDANSVCGHIRSVVSRLFGPEAAEAIQYGGSVKPDNIRDFLAQQQIDGPLVGGASL
161 171 181 191 201 211 221 231



[1/1] 21/09/2026, 22:47:07



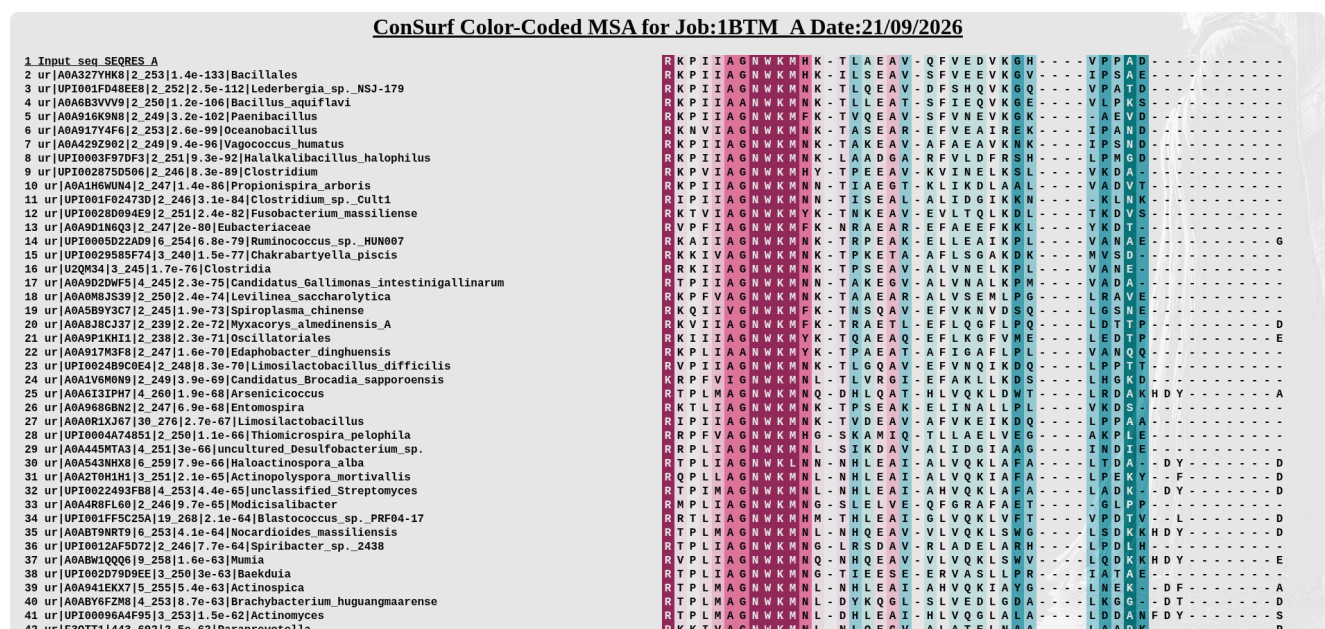


Insufficient Data

- [Download Scale](#)



this site gives lot of information they used neural nets as well



downloaded and added in the my repo (github)
named as `1BTM_A_consurf_grades.txt`

```
from -1.324 to -1.030 the grade is 9
from -1.030 to -0.736 the grade is 8
from -0.736 to -0.441 the grade is 7
from -0.441 to -0.147 the grade is 6
from -0.147 to 0.353 the grade is 5
from 0.353 to 1.059 the grade is 4
from 1.059 to 1.764 the grade is 3
from 1.764 to 2.470 the grade is 2
from 2.470 to 3.176 the grade is 1
```

- POS: The position of the acid in the sequence.
- SEQ: The acid one letter.
- ATOM: When there's a model, The ATOM derived sequence in three letter code, including the acid's positions as they appear in the model.
- SCORE: The normalized conservation scores.
- COLOR: The color scale representing the conservation scores (9 - conserved, 1 - variable).
- CONFIDENCE INTERVAL: When using the bayesian method for calculating rates, a confidence interval is assigned to each rate.
- B/E: Buried (b) or Exposed (e) residue.
- F/S: functional (f) or structural (s) residue (f - highly conserved and exposed, s - highly conserved and buried).
- MSA DATA: The number of aligned sequences having an acid (non-gapped) from the overall number of sequences at each position.
- RESIDUE VARIETY: The residues variety at each position of the multiple sequence alignment.

POS	SEQ	ATOM	SCORE	COLOR	CONFIDENCE	INTERVAL	B/E	F/S	MSA DATA	RESIDUE VARIETY
1	R	ARG:1:A	-1.276	9	-1.331,	-1.249 9,9	e	f	89/150	R 98%, K 1%
2	K	LYS:2:A	-0.239	6	-0.459,	-0.071 7,5	e		102/150	K 42%, R 29%, T
3	P	PRO:3:A	0.195	5	-0.071,	0.373 5,4	e		117/150	P 52%, K 20%, T
4	I	ILE:4:A	-0.263	6	-0.459,	-0.183 7,6	e		132/150	L 47%, I 27%, F
5	I	ILE:5:A	-0.554	7	-0.734,	-0.459 7,7	b		135/150	I 49%, V 32%, M
6	A	ALA:6:A	-0.751	8	-0.897,	-0.673 8,7	e	f	141/150	A 67%, G 12%, V
7	G	GLY:7:A	-0.968	8	-1.075,	-0.897 9,8	b		141/150	G 87%, A 7%, T
8	N	ASN:8:A	-1.299	9	-1.331,	-1.279 9,9	e	f	141/150	N 95%, S 4%, T
9	W	TRP:9:A	-1.168	9	-1.249,	-1.113 9,9	b	s	141/150	W 95%, F 3%, M
10	K	LYS:10:A	-1.322	9	-1.341,	-1.306 9,9	e	f	142/150	K
11	M	MET:11:A	-1.262	9	-1.306,	-1.249 9,9	e	f	143/150	M 96%, L 2%, A
12	H	HIS:12:A	-0.885	8	-0.990,	-0.846 8,8	e	f	143/150	N 77%, H 13%, Y
13	K	LYS:13:A	-0.358	6	-0.536,	-0.285 7,6	e		143/150	G 45%, K 24%, L
14	T	THR:14:A	-0.159	6	-0.376,	-0.071 6,5	e		143/150	T 39%, S 16%, N
15	L	LEU:15:A	1.115	3	0.579,	1.191 4,3	e		143/150	L 24%, R 16%, H
16	A	ALA:16:A	0.544	4	0.203,	0.579 5,4	e		143/150	A 34%, L 11%, S
17	E	GLU:17:A	-0.210	6	-0.376,	-0.071 6,5	e		142/150	E 42%, S 15%, A
18	A	ALA:18:A	-0.530	7	-0.673,	-0.459 7,7	b		142/150	A 44%, L 14%, M
19	V	VAL:19:A	1.065	3	0.579,	1.191 4,3	e		142/150	A 20%, V 12%, I
20	Q	GLN:20:A	0.893	4	0.373,	1.191 4,3	e		143/150	E 29%, A 26%, Q
21	F	PHE:21:A	-0.120	5	-0.376,	0.056 6,5	b		144/150	L 54%, F 20%, A
22	V	VAL:22:A	0.459	4	0.203,	0.579 5,4	b		144/150	V 29%, L 20%, I
23	E	GLU:23:A	0.568	4	0.203,	0.840 5,4	e		145/150	A 21%, Q 17%, E
24	D	ASP:24:A	1.056	4	0.579,	1.191 4,3	e		145/150	G 17%, E 15%, K
25	V	VAL:25:A	0.056	5	-0.183,	0.203 6,5	b		145/150	L 46%, I 17%, V
26	K	LYS:26:A	0.712	4	0.373,	0.840 4,4	e		145/150	A 26%, K 20%, L
27	G	GLY:27:A	2.033	2	1.191,	1.732 3,3	e		145/150	A 18%, S 11%, D
28	H	HIS:28:A	1.356	3	0.840,	1.732 4,3	e		145/150	A 24%, G 15%, I
29	V	VAL:29:A	1.111	3	0.579,	1.191 4,3	b		103/150	L 40%, V 19%, I
30	P	PRO:30:A	1.821	2	1.191,	1.732 3,3	e		136/150	P 16%, A 15%, K
31	P	PRO:31:A	1.650	3	1.191,	1.732 3,3	e		145/150	D 21%, L 15%, K